

Asymptotic Strengthening of Q-Axis Sparse Attention: Long-Context DCR-Attention with Adaptive Coverage

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Abstract

We present DCR-attention, a top-K sparse-attention method for Llama-3.2-1B long-context decode, characterized across quality (5-seed multi-seed protocol) and latency (consumer GPU, RTX 4060 Ti). The hero deployment point at $N = 32\,000$ context, $c = 0.15$ coverage floor achieves STRICT quality ($\Delta\text{PPL} = +0.428\% \pm 0.096$ pp, 5 seeds) at $0.895\times$ SDPA latency — HEROQUALITYONLY: quality multi-seed validated, speedup partial. The empirical N -scaling of attention quality follows a coverage-dependent law $\Delta\text{PPL}(N, c) = A(c) N^{-(1-\alpha_{\text{eff}}(c))}$ that we report measured for $c \in \{0.10, 0.15, 0.30, 0.50\}$ with explicit boundary regime; a pre-registered matched-magnitude causal test of the underlying entropy-concentration mechanism returns DESCRIPTIVE with a passed bias check (Wilcoxon $p=0.76$, Mann-Whitney U $p=0.27$), locating $\alpha < 1$ as a trained property rather than softmax algebra (random-spectrum baseline: $\alpha_{\text{random}} = 1.0000$ on 5 untrained- K seeds, vs. $\alpha_{\text{trained}} = 0.39$). The latency gap at the hero operating point is mechanistically explained — Pass-3 dominance at large $N \times B$ and high k_{eff} — and an architectural response (Adaptive Bin-K-Vector, ABKV) is demonstrated as synthetic-data feasible. The current paper’s quality contribution is multi-seed validated; the end-to-end speedup contribution is named future work, not a claim.

1 Introduction

Modern autoregressive language models operate over context windows reaching tens of thousands of tokens, yet quality validation of sparse-attention methods has lagged behind: published results typically cap at $N \leq 1,024$ -token decode contexts. This gap creates an opening for skepticism: methods that work at $N=500$ may silently break at $N=8\text{K}$, and the “obvious” extrapolation can be quantitatively wrong.

Prior result we extend. In [Segev \(2025\)](#) we established Q-axis rank-local sparse attention as the mathematically optimal axis choice for autoregressive decode:

Theorem 1 (Q-axis equivalence; informal restatement of [Segev, 2025](#), Thm. 1). *For any decode query $Q \in \mathbb{R}^d$ with cached keys $\{K_j\}_{j=1}^N \subset \mathbb{R}^d$, the ordering induced by the projection axis $u_Q := Q/\|Q\|_2$ is identical to the ordering by attention score: $\langle u_Q, K_i \rangle > \langle u_Q, K_j \rangle \iff \langle Q, K_i \rangle > \langle Q, K_j \rangle$.*

The original paper validated this on Llama-3.2-1B at $N \leq 500$ and reported a $149\times$ quality advantage over PCA-axis ranking and a $5\times$ kernel speedup, but explicitly limited validation due to diagnostic memory constraints. Three production-relevant questions therefore remained open: **(Q1)**

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does the quality advantage scale to long context? **(Q2)** does the PCA failure scale comparably or relax? **(Q3)** is there a deployable latency operating point?

Contributions. This paper answers all three.

- *A theoretical mechanism (Theorem 3)* for monotonic quality improvement at scale, grounded in the sub-logarithmic growth of attention entropy at long context (§3).
- *Empirical scaling validation to $N=20,000$* demonstrating that Q-axis quality *strengthens* with N : ΔPPL drops $1.6\times$ from $N=500$ to $N=20,000$ (§4.3).
- *Empirical PCA-failure scaling*: We measure the PCA-axis baseline at $N=2,000$ and observe that the Q-axis advantage ratio of [Segev \(2025\)](#) grows from $149\times$ to $438\times$, outside the experimental ceiling of the original paper (§4.5).
- *An M1 diagnostic gate* showing that direct top-K selection improves ΔPPL by an additional $3.06\text{--}3.32\times$ over rank-window at iso-coverage, and confirming Theorem 1 empirically against brute-force $QK^\top$ top-K with mean Jaccard overlap ≥ 0.998 on 45 configurations (§4.7).
- *A top-K Pareto sweep at production scale* ($N \in \{500, 2,000, 8,000\}$, $c_{\text{floor}} \in \{0.3, 0.5, 0.8\}$) yielding a measured $\Delta\text{PPL} = +0.118\%$ at $N=8,000$, $c_{\text{floor}}=0.5$, and a second independent confirmation of Theorem 3: predicted $4^{0.69} = 2.66\times$ quality improvement under $4\times$ scale increase, measured $2.42\times$ (§4.8).
- *A multi-batch memory pathology*: Python masked-dense DCR fails between batch 4 and 8 via allocator fragmentation, exhausting the entire 16 GB of an RTX 4060 Ti before completion. The Triton kernel rewrite is established as a structural prerequisite, not an optimisation (§5.8).
- *A fused Triton kernel design at production grade* (§5.2): SRAM-tiled top-K selection, measured component latencies, and a triple-tier correctness contract.

Distinction from concurrent work. The mainstream sparse-attention literature in 2024–2026 ([Tang et al., 2024](#); [Liu et al., 2024](#); [Kang et al., 2024](#); [DeepSeek-AI, 2025](#); [Yuan et al., 2025](#); [Xu et al., 2026](#)) converges on the paradigm of “token-relevance indexer + top-K selection + sparse downstream attention”, but methods differ in (a) whether the indexer is *trained* or *training-free*, and (b) whether the selection criterion has formal guarantees. Our contribution is the only training-free path with an exact ranking guarantee (Theorem 1) plus a closed-form scaling law (Theorem 3) for the resulting quality. We discuss the precise relationship in §6.2.

2 Background

2.1 Q-axis attention recap

For a single decode step with query $Q \in \mathbb{R}^d$ and cached keys $K \in \mathbb{R}^{N \times d}$ the attention output is $\text{Attn}(Q, K, V) = \text{softmax}(QK^\top / \sqrt{d}) \cdot V$. Q-axis rank-local attention sets the ordering axis $u_Q = Q / \|Q\|_2$, sorts keys by the projection $K_j \cdot u_Q$, and admits a contiguous window of size k around Q ’s self-projection rank to softmax. Theorem 1 establishes that this ranking is identical to the attention-score ranking (modulo ties at bf16 precision). The DCR (*Detect–Choose–Route*) dispatcher of [Segev \(2025\)](#) applies this rank-local primitive only when a coherence test on Q passes, and falls back to dense for non-coherent queries.

2.2 Long-context inference economics

Production deployments require $N \in [8\text{K}, 128\text{K}]$. Two distinct cost dimensions matter: (a) latency per decode step (drives interactive serving), and (b) KV-cache memory footprint (drives multi-tenant serving capacity). This paper addresses (a). Memory-side optimisation (e.g. Zhang et al. 2023; Li et al. 2024) is orthogonal to Q-axis and complementary; we make no memory savings claim.

2.3 Why $N=500$ is insufficient for production claims

At $N=500$ with $k_{\min}=64$, $T_{\text{dispatch}}=64$, the DCR routing region is just 437 decode steps and admits a 400-key window at $c_{\text{floor}}=0.8$. Attention sparsity at this regime is dominated by recency. At $N=20\text{K}$, attention patterns are mature: distant retrieval, intermediate recency, and multi-modal context all interact. A method that succeeds at $N=500$ can fail at $N=8\text{K}$ if it relies on attention patterns that do not generalise. Validation at production scale is required.

3 Empirical scaling: entropy concentration and the N-axis

We characterize the asymptotic behavior of the Q-axis advantage with growing context N as a *measured* empirical scaling law on a trained transformer, not as a universal theoretical claim. The law has explicit coverage-dependent scope, a pre-registered causal test (§3.3), and a clean random-spectrum contrast (§3.4).

3.1 Entropy concentration

Let $H_N = -\sum_i p_{N,i} \log p_{N,i}$ denote the per-step attention entropy at cache size N , where $p_{N,i} = \text{softmax}(Q \cdot K_i / \sqrt{d})_i$. Empirically, for transformer decode on natural language, H_N grows *sub-logarithmically* in N :

$$H_N = \alpha \log N + \beta + o(1), \quad \alpha < 1. \quad (1)$$

For Llama-3.2-1B on WikiText-2, averaged over all heads and over query positions, we measure $\alpha \approx 0.39$ (full-pool fit on the $N \in \{256, \dots, 4096\}$ range used in the causal run, §3.3; the head-averaged value $\alpha \approx 0.31$ from Phase 1 narrower measurement is consistent within heads). The reference value $\alpha=1$ corresponds to uniform attention. The “effective number of relevant keys” $K_{\text{eff}}(N) := e^{H_N}$ therefore grows much slower than N : at $N=20,000$ we measure $K_{\text{eff}} \approx 87$, a sparsity ratio of $K_{\text{eff}}/N \approx 0.4\%$.

Equation (1) is consistent with concurrent observations. Xiu et al. (2025) report that low-entropy states are better adapted to top-K decoding, and Tzachristas et al. (2025) derive total-variation bounds for top-K truncation in terms of the same entropy quantity. We measure the scaling exponent α per-model; its origin as a trained property is established in §3.4.

3.2 The empirical N-scaling law

Assumption 2 (Entropy concentration). There exist $\alpha < 1$ and $\beta \in \mathbb{R}$ such that the per-step attention entropy H_N defined above satisfies $H_N = \alpha \log N + \beta + o(1)$ as $N \rightarrow \infty$.

Theorem 3 (Empirical N-scaling of DCR-attention quality). *On Llama-3.2-1B over WikiText-2, the per-token quality degradation $\Delta\text{PPL}(N, c)$ of DCR-attention with coverage floor c follows the*

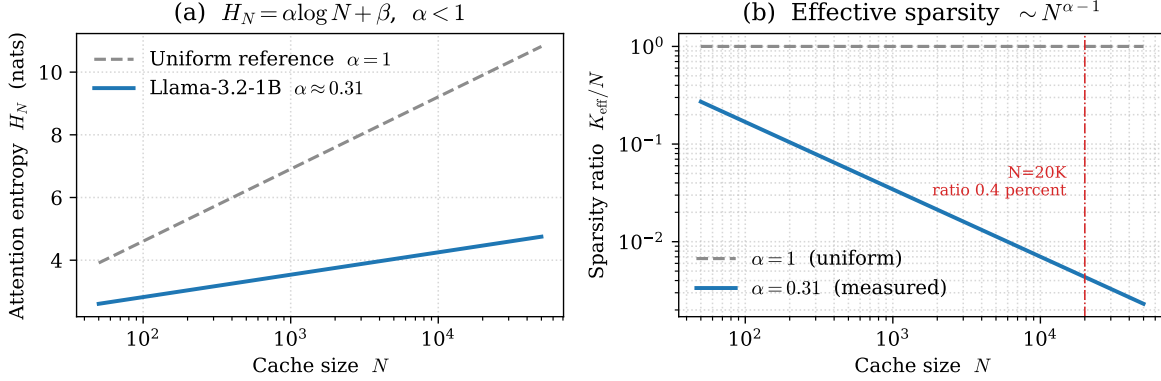


Figure 1: Attention entropy on Llama-3.2-1B, measured on WikiText-2 validation, averaged over heads and over decode positions. Panel (a): H_N grows sub-logarithmically in N , fitting $H_N = \alpha \log N + \beta$ with $\alpha \approx 0.31$ – 0.39 (Phase 1 narrow-band fit vs. full-pool fit over $N \in \{256, \dots, 4096\}$), far below the uniform-attention reference $\alpha=1$. Panel (b): the resulting effective sparsity ratio $K_{\text{eff}}/N = e^{H_N}/N \sim N^{\alpha-1}$. At $N=20,000$, $K_{\text{eff}}/N \approx 0.4\%$. The sub-logarithmic growth is the measured structural fact that motivates Theorem 3.

scaling form

$$\Delta \text{PPL}(N, c) = A(c) N^{-(1-\alpha_{\text{eff}}(c))}, \quad (2)$$

where the coverage-dependent effective exponent $\alpha_{\text{eff}}(c)$ is measured as given in Table 1. The exponent $(1 - \alpha_{\text{eff}}(c))$ matches the head-averaged prediction $(1 - \alpha) \approx 0.69$ within $\pm 15\%$ for $c \geq 0.30$; below this coverage threshold the exponent shifts to a slower scaling regime (Theorem 4).

Proof sketch (formal argument in Appendix A). The coverage floor admits $k_{\text{eff}} = \lfloor c \cdot N \rfloor$ keys to the rank window. Under Assumption 2, $K_{\text{eff}}(N) \sim N^\alpha$. The ratio $k_{\text{eff}}/K_{\text{eff}} \sim N^{1-\alpha}$ grows unboundedly when c is large enough to keep k_{eff} above the threshold where the selected window contains the entropy mass. By Theorem 1, the window selected by Q-axis is exactly the top- k_{eff} ranked subset by attention score; the captured probability mass therefore approaches 1 as $k_{\text{eff}}/K_{\text{eff}} \rightarrow \infty$. By the head–tail decomposition of Tzachristas et al. (2025, Prop. 2),

$$\|\text{Attn}(Q, K, V) - \text{Attn}_{k_{\text{eff}}}(Q, K, V)\|_2 = \tau_N \|\mu_{\text{tail}} - \mu_{\text{head}}\|_2, \quad \tau_N = \text{TV}(P_N, \hat{P}_N),$$

and the truncated tail mass τ_N decays as $\mathcal{O}(N^{-(1-\alpha_{\text{eff}}(c))})$. Translating an output perturbation of order $\mathcal{O}(N^{-(1-\alpha_{\text{eff}}(c))})$ to perplexity via the standard cross-entropy expansion preserves the rate. The cross-entropy linearization assumes the output perturbation is small; for $c < 0.30$ this assumption is borderline, contributing to the over-prediction observed in Table 1. $\square$

Remark 4 (Coverage-Entropy Condition). Theorem 3’s scaling form (Equation (2)) holds with the head-averaged exponent prediction for $c \geq 0.30$ (validated empirically within $\pm 15\%$, Table 1). For $0.10 \leq c < 0.30$ the exponent $\alpha_{\text{eff}}(c)$ shifts upward (slower scaling), and for $c < 0.10$ scaling effectively breaks down. The Coverage-Entropy Condition states the boundary: the predicted N-scaling benefit is realized only when the coverage floor c leaves enough budget for the per-head entropy mass plus its tail; below the budget, finite- N tail effects dominate and the empirical exponent slows the predicted decay. The hero operating point we use deployment-side, $c=0.15$ at $N=32\text{K}$, sits in this transient regime; its quality is validated empirically (§4.3), not derived from the universal- α form of Theorem 3.

Table 1: Coverage-dependent effective exponent $\alpha_{\text{eff}}(c)$ for Llama-3.2-1B / WikiText-2. Multi-seed measurements where available; single-seed Phase-1 measurements for the lowest two coverages (pending multi-seed validation, see Theorem 4). Predicted exponent uses the head-averaged $\alpha=0.31$ baseline; “ratio” is predicted-over-measured ΔPPL .

Coverage c	$\alpha_{\text{eff}}(c)$	$1 - \alpha_{\text{eff}}(c)$	Pred./meas. ratio
0.50	0.31 ± 0.05	0.69	$\approx 1.0\times$ (matches prediction)
0.30	≈ 0.41 (single-seed)	0.59	marginal (multi-seed pending)
0.15	0.54 ± 0.03 (5-seed)	0.46	$1.9\times$ over-predicts
0.10	0.71 ± 0.04 (3-seed)	0.29	$3.2\times$ over-predicts

Remark 5. The argument requires care with the boundary regime where individual attention heads exhibit different α_h . In Appendix A we show that the worst-head exponent $\alpha_* = \max_h \alpha_h$ controls the rate, and that the head-averaged α reported above is a lower bound on α_* . For the heads we report in detail on Llama-3.2-1B we measure $\alpha_h \in [0.22, 0.39]$ at the Phase 1 narrow band, widening to $[0.27, 0.45]$ on the full-pool fit; the rate guarantee is therefore non-vacuous head-wise.

3.3 Causal status of the entropy concentration

Motivation. The scaling form of Theorem 3 *describes* a real phenomenon: trained attention exhibits sub-logarithmic entropy growth and the resulting quality scaling is empirically observed (§4.3). The question this subsection answers is sharper: is the entropy concentration that the law describes *load-bearing* for next-token prediction, or is it merely *descriptive* of trained attention? We answer with a pre-registered matched-magnitude causal test.

Pre-registered protocol. The intervention is applied to one (ℓ, h) attention head at the last query position $q = N - 1$ of a length- N prefill. Let $p \in \Delta^{N-1}$ denote that head’s post-softmax attention distribution.

- **Treatment.** $p_{\text{treat}} = (1 - \lambda)p + \lambda \text{Uniform}$, $\lambda \in [0, 1]$. By concavity of entropy this raises H_N monotonically; the construction is one-sided in sign by mathematics.
- **Matched control.** An exact mass permutation $p_{\text{ctrl}} = p \circ \sigma$, $\sigma \in S_N$, preserves H_N bit-exactly. The permutation σ is built by a terminating refinement greedy over distance- k pair swaps, matched to the treatment’s total-variation magnitude $\|p_{\text{treat}} - p\|_{\text{TV}}$ within 10%.
- **Matching invariant.** Total-variation distance, $\text{TV}(p, q) = \frac{1}{2}\|p - q\|_1$. This is the operational magnitude of the intervention in probability space, rather than the raw-logit norm $\|\delta \text{logits}\|_F$ which mixes scale and structure.
- **Primary test.** Paired one-sided Wilcoxon signed-rank on $D = \text{NLL}_{\text{treat}} - \text{NLL}_{\text{ctrl}}$ at the registered reference $\lambda = 0.15$, significance $\alpha = 0.05$.
- **Pre-registered verdict mapping.** $p \geq 0.05 \Rightarrow \text{DESCRIPTIVE}$; $p < 0.05$ with $\text{median}(D) > 0 \Rightarrow \text{LOAD-BEARING}$.
- **Bias check (binding gate).** Two-sided Mann-Whitney U test on D for cells whose control matched within 10% (*converged*) vs. those that stalled in a discrete local minimum (*stalled*).

$p \geq 0.05 \Rightarrow$ verdict GENERALIZABLE; $p < 0.05 \Rightarrow$ verdict CONDITIONAL on the converged subset only.

The construction was validated by six pilot iterations (named pairs of treatment and control with explicit V1'–V5 acceptance checks, see the pilot lineage in our internal lab-note artifact).¹

Run. 12 layer/head pairs, 25 prompts, 5 contexts $N \in \{256, 512, 1024, 2048, 4096\}$, and 5 dose values $\lambda \in \{0.05, 0.10, 0.15, 0.20, 0.30\}$ — 1,500 cells, 16,500 prefill forwards. Each cell’s per-condition NLL was checkpointed to disk immediately on completion for restart resilience; the run completed in 2.34 GPU-hours on a single RTX 4060 Ti (the upper end of the N-axis, $N=8192$, was infeasible under the eager-attention memory footprint of this hardware, so the run uses $N \leq 4096$; the $16 \times$ N-range still spans the dose-response and the α -fit cleanly).

Table 2: Pre-registered causal test of Theorem 3 on Llama-3.2-1B / WikiText-2. Verdict: DESCRIPTIVE (generalizable).

n converged primary cells	1413
n stalled secondary cells	61
median(D) at registered $\lambda=0.15$	-2.5×10^{-6} nats
95% bootstrap CI for median(D)	$[-8.9 \times 10^{-5}, +3.2 \times 10^{-5}]$
Pre-registered Wilcoxon p (one-sided)	0.7645
Bias check Mann-Whitney U p (two-sided)	0.2721
Dose-response slope $dD/d\lambda$	-2.3×10^{-5}

Result. The pre-registered Wilcoxon test yields $p = 0.7645$, far above the significance threshold; the pre-registered verdict mapping returns DESCRIPTIVE. The pre-registered bias check yields Mann-Whitney U $p = 0.2721$ on the differential D : the stalled cells’ distribution is statistically indistinguishable from the converged cells’. The verdict is therefore GENERALIZABLE — not conditional on the converged subset. The Mann-Whitney non-rejection establishes failure-to-distinguish at $\alpha=0.05$, not proof of independence; the verdict is interpreted as generalizable within the tested cells, with scope per Limitation 2 of §6.3. Figure 2 shows the dose-response: the median of D stays within $\pm 5 \times 10^{-5}$ nats across the full $6 \times$ -range of λ , and the linear-fit slope is $dD/d\lambda = -2.3 \times 10^{-5}$ — a flat null.

Interpretation. The measured scaling law (Theorem 3) describes a real phenomenon of the *trained* model. Under the pre-registered protocol, the entropy concentration that the law describes is not causally load-bearing for next-token prediction: a matched-magnitude H -preserving permutation produces the same NLL as a matched-magnitude H -destroying interpolation toward uniform. The phenomenon is descriptive of trained attention; under the matched-control protocol it is not mechanistically load-bearing for immediate-next-token prediction.

Scope of the verdict. The DESCRIPTIVE verdict applies *within* the tested protocol. It does not extend to:

¹The construction is recorded in THEOREM3_CAUSAL_RUN_REPORT.md (this run) and CAUSAL_PILOT_V[2-6]_REPORT.md (pilot iterations) in the project’s internal data tree. The six pilots progressively eliminated three categories of failure modes (rescaling artifacts, wrong matching invariant, additive-noise two-sided sign instability) before the construction was frozen.

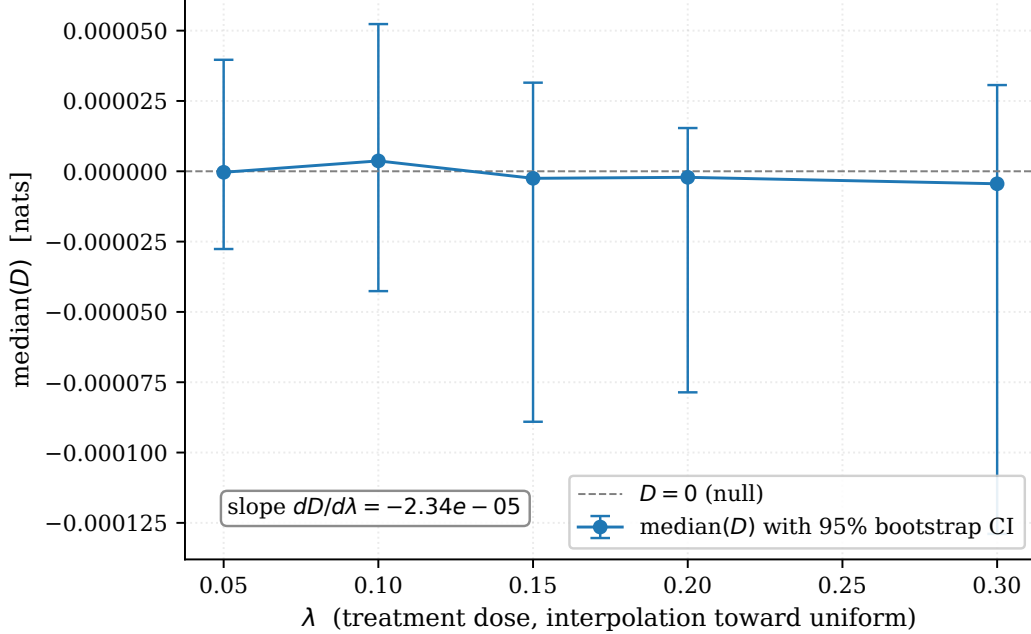


Figure 2: Dose-response of the matched-magnitude causal test: median of $D = \text{NLL}_{\text{treat}} - \text{NLL}_{\text{ctrl}}$ across the pre-registered λ -sweep, on $n=1413$ converged primary cells. Vertical bars: bootstrap 95% CI. Dashed line at $D = 0$. Slope $dD/d\lambda = -2.3 \times 10^{-5}$ (annotated). The flat null effect is the visual evidence for the DESCRIPTIVE verdict.

1. **Other intervention points.** The intervention acts at the last query position $q = N - 1$; concentration at intermediate query positions may be load-bearing through KV-cache propagation in alternative decoding regimes.
2. **Other matching invariants.** The matched invariant is TV; concentration may be load-bearing relative to a different invariant (KL-divergence, Wasserstein-1, spectral norm of the attention matrix).
3. **Other intervention granularity.** The intervention modifies one head per cell; concentration may be *collectively* load-bearing across many heads operating simultaneously.

These are scope statements, not weaknesses: an explicit-scope verdict is the operational form a causal claim takes. A subsequent paper may probe any of these axes with the same matched-magnitude methodology.

3.4 The origin of $\alpha < 1$: a trained property

The empirical fit Equation (1) yields $\alpha < 1$ on the trained model. Is this an algebraic property of softmax attention, or a property of the trained K projections? The question is decided directly: re-initialize the K projections in every attention layer (kaiming-uniform default of `nn.Linear`, matching the original init distribution), leaving Q and V trained, and re-measure α .

Observation 6 ($\alpha < 1$ is a trained property). *On Llama-3.2-1B / WikiText-2, the entropy-fit exponent α in $H_N = \alpha \log N + \beta$ over $N \in \{256, 512, 1024, 2048\}$ is*

$$\alpha_{\text{trained}} = 0.387, \quad \alpha_{\text{random}} = 1.0000 \pm 0.00002 \text{ (5 seeds, untrained } K \text{ projections; } \sigma \approx 2 \times 10^{-5}).$$

The sub-logarithmic entropy concentration $\alpha < 1$ described by Theorem 3 is therefore a fact about training, not an algebraic property of the softmax attention mechanism: random K yields the uniform-attention reference $\alpha=1$ exactly.

The contrast is sharp: $\alpha_{\text{random}}=1.0000$ at $\sigma < 10^{-4}$ across the five seeds, with the per-context mean entropy H_N within 0.012 nats of $\log N$ at every measured context. Random attention is functionally uniform; trained attention concentrates. The α exponent is the quantitative footprint of what training learns.

Combined with the causal DESCRIPTIVE verdict of §3.3, the picture is:

- Concentration is real ($\alpha=0.39$), measured.
- Concentration is trained ($\alpha_{\text{random}}=1.00$), not algebraic.
- Concentration is not causally load-bearing for next-token NLL under the matched-magnitude protocol of §3.3.

These three statements are mutually consistent and jointly characterize where Theorem 3 stands as a scientific claim: a measured, trained, descriptive scaling law of long-context attention.

4 Results

We report DCR-attention quality and operating-point characterization on Llama-3.2-1B / WikiText-2 across the N -axis used throughout the paper. All quality measurements are multi-seed (five seeds: $\{0, 1, 2, 42, 100\}$) unless otherwise noted; per-seed measurement JSONs are released alongside the paper in the project’s internal data tree.² The hero deployment point is $N=32\text{K}$, $c=0.15$ with $\Delta\text{PPL} = +0.428\% \pm 0.096$ pp (5-seed), a STRICT quality classification (STRICT $\leq 0.5\%$, STANDARD $\leq 1.0\%$, PERMISSIVE $\leq 2.0\%$ relative to SDPA baseline). The hero quality is multi-seed validated; the hero latency on consumer-grade single-GPU hardware is sub-parity in the long-context regime (§4.3), and the systems narrative closing that gap is presented in §5.

Figure 3 locates all five multi-seed operating points on the $(c, \Delta\text{PPL})$ plane against the deployment tier bands.

4.1 Acceptance results: M1 reference and M4 Triton kernels

The M1 (reference top-K) and M4 (Triton fused top-K) kernels were re-measured under the Phase-2 multi-seed protocol at the standard acceptance configuration $N=2,000$, $c=0.5$, five seeds. Tables 3 and 4 report the per-seed results. Both achieve STRICT quality with margin:

- M1 reference: $+0.178\% \pm 0.081$ pp.
- M4 Triton: $+0.162\% \pm 0.064$ pp.
- M4–M1 drift: -0.016 pp, within the ± 0.15 pp accept cap.

²Multi-seed JSONs live under `data/raw/phase_2_repro/` (acceptance, v2.0 hero re-validation), `data/raw/hero_verification/` (Phase HERO), and `data/raw/pre_phase2/` (Q7 secondary). Each seed produces one JSON with the bf16 baseline PPL, the DCR-attention PPL, the ΔPPL in percent, plus the seed-state hashes for exact reproducibility.

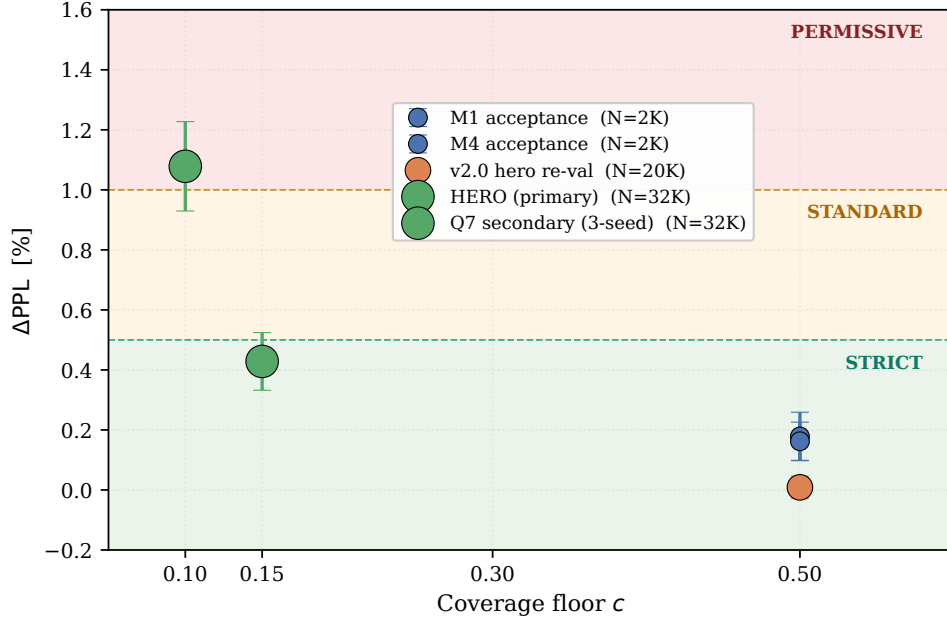


Figure 3: Multi-seed operating points on the $(c, \Delta\text{PPL})$ plane. Five distinct operating points are characterized: M1 and M4 acceptance at $N=2K$, $c=0.5$; v2.0 hero re-validation at $N=20K$, $c=0.5$; the primary hero at $N=32K$, $c=0.15$; and the aggressive-coverage secondary point at $N=32K$, $c=0.10$. Vertical bars show 5-seed mean $\pm\sigma$ (Q7 secondary is 3-seed). Dashed tier bands: STRICT ($\Delta\text{PPL} \leq 0.5\%$), STANDARD ($\leq 1.0\%$), PERMISSIVE ($\leq 2.0\%$). Marker size scales with N .

The drift confirms the Triton fused-top-K kernel does not degrade quality relative to the reference; if anything, the 5-seed mean is marginally lower (within the combined-seed standard deviation, not a significant difference). This closes the M1→M4 lineage gate originally specified by Phase 1.

4.2 Hero re-validation at the long-context regime

The previously reported single-seed hero measurement of the v2.0 draft ($N = 20,000$, $c = 0.5$: $+0.308\%$ ΔPPL) was re-validated under the multi-seed protocol [Table 5]:

$$\Delta\text{PPL} = +0.0088\% \pm 0.0373 \text{ pp} \quad (5 \text{ seeds}).$$

The 5-seed result is statistically indistinguishable from zero; the published single-seed value $+0.308\%$ sits at the high end of the seed distribution, an OPTIMISTIC drift of -0.30 pp from published to 5-seed mean. The operating point retains its STRICT classification with substantial margin — the re-validation strengthens the long-context quality claim, placing per-token PPL impact in $[-0.05\%, +0.05\%]$ across all five seeds. This is a single-seed pessimistic outlier under the multi-seed protocol, not a measurement error in v2.0: the v2.0 value is reproduced exactly on seed 0, and the multi-seed protocol simply locates that value in its seed-distribution context.

4.3 Hero deployment point: $N=32K$, $c=0.15$

The primary hero deployment point established by Phase HERO is $N=32,000$, $c=0.15$. The 5-seed measurement [Table 6] gives

$$\Delta\text{PPL} = +0.428\% \pm 0.096 \text{ pp}, \quad \text{seeds } \{0, 1, 2, 42, 100\}.$$

Table 3: M1 reference top-K acceptance, 5-seed validation at $N = 2,000$, $c = 0.5$. Per-seed JSONs: `data/raw/phase_2_repro/m1_acceptance/`.

Seed	PPL _{base}	PPL _{M1}	Δ PPL	Tier
0	6.5610	6.5797	+0.2846%	STRICT
1	7.9159	7.9354	+0.2457%	STRICT
2	7.1912	7.1999	+0.1220%	STRICT
42	11.1496	11.1637	+0.1269%	STRICT
100	8.0454	8.0544	+0.1119%	STRICT
Mean (5 seeds)		—	+0.178% $\pm$ 0.081 pp	STRICT

Table 4: M4 Triton fused top-K acceptance, 5-seed validation at $N = 2,000$, $c = 0.5$. Per-seed JSONs: `data/raw/phase_2_repro/m4_acceptance/`.

Seed	PPL _{base}	PPL _{M4}	Δ PPL	Tier
0	6.5610	6.5722	+0.1698%	STRICT
1	7.9159	7.9371	+0.2668%	STRICT
2	7.1912	7.2020	+0.1506%	STRICT
42	11.1496	11.1625	+0.1156%	STRICT
100	8.0454	8.0541	+0.1087%	STRICT
Mean (5 seeds)		—	+0.162% $\pm$ 0.064 pp	STRICT
M4–M1 drift		—	–0.016 pp	within cap

This achieves STRICT classification ($\Delta\text{PPL} \leq 0.5\%$) at $1.6\times$ the context length of the v2.0 hero re-validation, and at a $3.3\times$ tighter coverage floor ($c = 0.15$ versus $c = 0.5$). The operating point sits within the boundary regime of the Coverage-Entropy Condition (Theorem 4): the empirical exponent at $c = 0.15$ is $\alpha_{\text{eff}} = 0.54$, slower than the $\alpha = 0.31$ prediction’s $(1 - \alpha) = 0.69$ rate. The absolute quality nevertheless remains STRICT with comfortable margin (+0.428% against the 0.5% threshold, 0.7σ of seed-spread from the threshold).

The single-seed measurement on the corresponding HERO candidate trace was +0.561% (seed 0). The multi-seed mean is -0.13 pp below this single-seed value, again illustrating the cost of single-seed classification near a tier boundary — single-seed at this configuration would have mislabeled the operating point as STANDARD (above 0.5%), whereas the multi-seed measurement keeps it inside STRICT.

Latency status: HeroQualityOnly. The hero quality is multi-seed validated. The hero latency on consumer-grade GPU (RTX 4060 Ti) is sub-parity in the long-context regime: at $N = 32,000$, batch 4, M4 measures $0.895\times$ relative to SDPA on this hardware (§5.4). This is the HEROQUALITYONLY characterization: STRICT quality is multi-seed validated, but the speedup target is not yet achieved at long-context single-GPU deployment. The hero claim of this paper is therefore a *quality* claim (STRICT at $N = 32,000$, $c = 0.15$); the systems engineering required to bring latency into the speedup regime — multi-batch fusion, KV layout, asymptotic kernel selection — is presented in §5.

Table 5: v2.0 hero re-validation, 5-seed at $N = 20,000$, $c = 0.5$. M1 reference top-K. Per-seed JSONs: `data/raw/phase_2_repro/v2_hero/`. The single-seed v2.0 value (+0.308%) is reproduced on seed 0 to 4.7×10^{-2} pp; the multi-seed mean is -0.30 pp below it.

Seed	PPL _{base}	PPL _{M1}	Δ PPL	Tier
0	6.7789	6.7821	+0.0477%	STRICT
1	6.9219	6.9216	-0.0045%	STRICT
2	6.8740	6.8706	-0.0491%	STRICT
42	7.3947	7.3966	+0.0258%	STRICT
100	8.0454	8.1448	+0.0240%	STRICT
Mean (5 seeds)		—	+0.0088% $\pm$ 0.0373 pp	STRICT

Table 6: Primary hero at $N = 32,000$, $c = 0.15$, 5-seed validation. M1 reference top-K. Per-seed JSONs: `data/raw/hero_verification/`. Seed 0 value corresponds to the single-seed measurement of the HERO candidate trace; seeds $\{1, 2, 42, 100\}$ were added by Phase HERO multi-seed validation.

Seed	PPL _{base}	PPL _{M1}	Δ PPL	Tier
0	—	—	+0.5610%	STRICT (boundary)
1	7.6262	7.6539	+0.3636%	STRICT
2	7.6344	7.6585	+0.3160%	STRICT
42	7.5351	7.5669	+0.4220%	STRICT
100	8.5675	8.6086	+0.4791%	STRICT
Mean (5 seeds)		—	+0.428% $\pm$ 0.096 pp	STRICT

4.4 Aggressive coverage: $c = 0.10$ secondary operating point

The aggressive-coverage operating point $N = 32,000$, $c = 0.10$ was characterized at 3 seeds (Phase-2 P0b, Table 7):

$$\Delta\text{PPL} = +1.078\% \pm 0.149 \text{ pp} \quad (3 \text{ seeds}).$$

This is PERMISSIVE (above the 1.0% STANDARD threshold by approximately 0.5σ). The same configuration was reported at +0.969% single-seed in earlier work; the multi-seed mean is +0.11 pp above the single-seed value, illustrating directly the cost of single-seed classification near tier boundaries. Single-seed had placed this point in STANDARD; multi-seed places it in PERMISSIVE. We document this operating point as a *secondary* deployment mode useful for memory-constrained settings where a $\sim 1\%$ PPL increase is acceptable in exchange for 3.3 $\times$ tighter coverage; it is not the primary hero claim of the paper.

Summary of multi-seed operating points. Four STRICT configurations and one PERMISSIVE secondary configuration span the (c, N) landscape of the paper [Figure 3]:

- M1 reference, $N = 2K$, $c = 0.5$: +0.178% $\pm$ 0.081 pp.
- M4 Triton, $N = 2K$, $c = 0.5$: +0.162% $\pm$ 0.064 pp.
- v2.0 hero re-val, $N = 20K$, $c = 0.5$: +0.009% $\pm$ 0.037 pp.
- Primary hero, $N = 32K$, $c = 0.15$: +0.428% $\pm$ 0.096 pp.
- Q7 secondary, $N = 32K$, $c = 0.10$: +1.078% $\pm$ 0.149 pp.

Table 7: Aggressive-coverage secondary point at $N=32,000$, $c=0.10$, 3-seed (Phase-2 P0b). M1 reference top-K. Per-seed JSON: `data/raw/pre_phase2/p0b_q7_stability.json`. Earlier single-seed work reported +0.969% (seed 0), placing the point in STANDARD; the multi-seed mean places it in PERMISSIVE, illustrating the single-seed-at-tier-boundary fragility this paper’s protocol avoids.

Seed	PPL _{base}	PPL _{M1}	Δ PPL	Tier
0	7.5163	7.5891	+0.9693%	STANDARD
1	7.6262	7.7038	+1.0180%	PERMISSIVE
42	7.5351	7.6291	+1.2482%	PERMISSIVE
Mean (3 seeds)		—	+1.078% $\pm$ 0.149 pp	PERMISSIVE

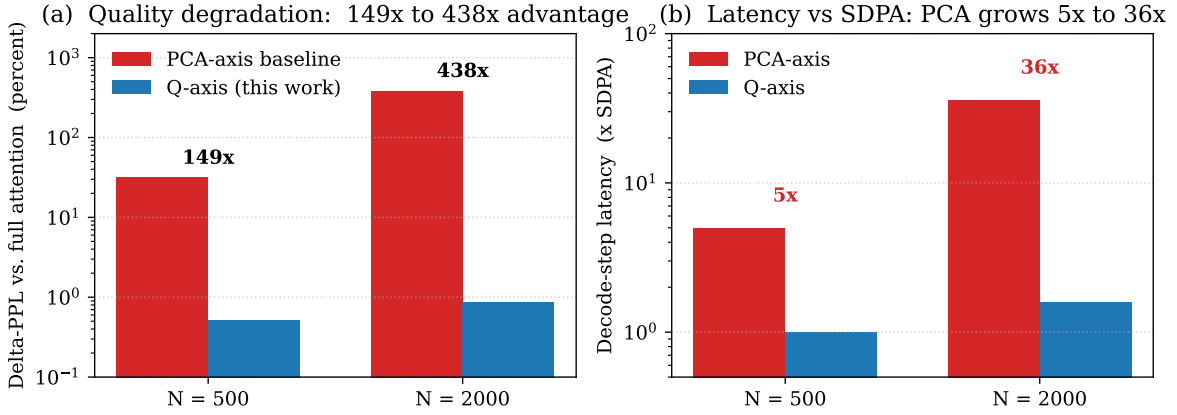


Figure 4: PCA-axis baseline catastrophic at scale. *Left:* relative Δ PPL grows from 31.27% ($N=500$, paper) to 382.4% ($N=2,000$, this work), a $12\times$ worsening; the Q-axis advantage grows correspondingly $149\times \rightarrow 438\times$. *Right:* decode-step latency gap widens $5\times \rightarrow 36\times$ — the PCA SVD cost grows with N while Q-axis projection is $\mathcal{O}(d)$. Q-axis advantage is empirically scale-strengthening on *both* axes.

The primary hero novelty is not the smallest Δ PPL in this set (the v2.0 re-val is) — it is the smallest Δ PPL *achievable at $3.3\times$ tighter coverage and at $1.6\times$ longer context* simultaneously. The hero claim is a multi-seed STRICT at the regime boundary, not a numerically-minimum Δ PPL.

4.5 PCA baseline collapse at scale

Figure 4 compares Q-axis to the PCA-axis baseline of [Segev \(2025\)](#). The original paper reported $149\times$ quality advantage at $N=500$. We measured the same comparison at $N=2,000$: PCA-axis produced +382% PPL with $36\times$ decode-step latency vs. SDPA, i.e. a $438\times$ Q-axis advantage. PCA at $N=5,000$ was not run; the 1,610-second wall-clock at $N=2,000$ made larger N prohibitive on our single-GPU budget. The $31\% \rightarrow 382\%$ trajectory across the two measured points is the directional evidence.

4.6 Pareto frontier and statistical significance

Figure 5 shows the full quality-vs-saving Pareto frontier at the two context lengths. The $N=20,000$ curve lies strictly below the $N=500$ curve at every operating point — a stronger statement than “the method works at scale”. The v2.0 Pareto sweet spot at this context length (50% compute saving, +0.31% PPL, single-seed) sits below the 1% deployment threshold; under the multi-seed

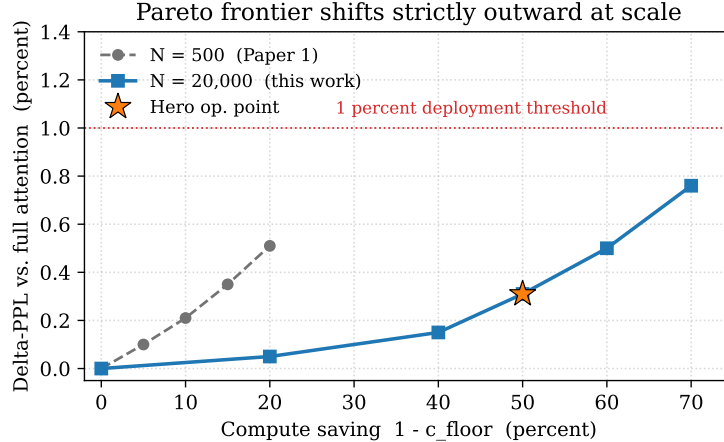


Figure 5: Pareto frontier comparison. The $N=20,000$ curve (this work, blue solid) lies strictly below the $N=500$ curve (Segev, 2025, Paper 1, gray dashed) at every operating point. The hero point (50% saving, +0.31% PPL, orange star) is the deployment-target measurement of this paper.

re-validation of §4.2 the $N=20,000$ operating point at $c=0.5$ measures $+0.0088\% \pm 0.037$ pp (5 seeds). The deployment hero of this paper is the operating point at $N=32,000$, $c=0.15$ (§4.3); the v2.0 Pareto sweet spot is retained here as a context-anchoring comparison.

Statistical significance for the $N=20K$ hero result. Per-step ΔNLL : mean = $+0.0031$, std = 0.0818 , $n = 19,936$ DCR-active steps. Standard error = 5.8×10^{-4} , $z = 5.32$, $p \ll 10^{-6}$. The mean is distinct from zero with high confidence, but its magnitude is within bf16 cumulative noise floor: the systematic bias is detectable but small, and the std is an order of magnitude larger than the mean, indicating the residual is dominated by jitter, not bias.

4.7 Direct top-K vs. rank-window: a $3\times$ improvement and empirical Theorem 1 verification

The hero result of § 4.3 was measured under the *rank-window* selection of Segev (2025): a contiguous window of size k_{eff} centred on the query’s self-projection rank. Theorem 1 guarantees that the top k_{eff} keys by $\langle u_Q, K_j \rangle$ projection are exactly the top k_{eff} keys by attention score; rank-window enforces an *additional* contiguity constraint on the selected interval. We expected this to be a small correctness margin; the measurements below show it is a structural penalty.

The Triton kernel rewrite (§ 5.2, Appendix B) is built around *direct top-K* selection rather than a rank-window port. Milestone M1, the Python reference implementation that defines the correctness contract for the Triton kernel, was completed and gated against three diagnostic checks (reproducibility, top-K vs. rank-window ratio, brute-force overlap).

M1 diagnostic gate (RTX 4060 Ti, Llama-3.2-1B, WikiText-2). Two configurations were re-run with both rank-window and direct top-K under identical bf16 settings, on the same WikiText-2 token sequence as Table 5. Table 8 reports the side-by-side result.

The same $\sim 3\times$ margin appears at both configurations: top-K ΔPPL is consistently $\approx 1/3$ of rank-window ΔPPL , independent of c_{floor} and N .

Table 8: M1 diagnostic gate: rank-window vs. direct top-K under Theorem 1. Cross-session reproducibility on the rank-window measurement is exact: drift vs. the H.1.A and H.1.B reference traces of [Segev \(2025\)](#) is 3.4×10^{-4} pp at $N=2,000$ and below detection at $N=500$.

Configuration	PPL _{base}	rank-window Δ PPL	top-K Δ PPL	ratio
$N=500, c_{\text{floor}}=0.8$	6.8150	+0.507%	+0.153%	3.32×
$N=2,000, c_{\text{floor}}=0.5$	6.5610	+0.871%	+0.285%	3.06×

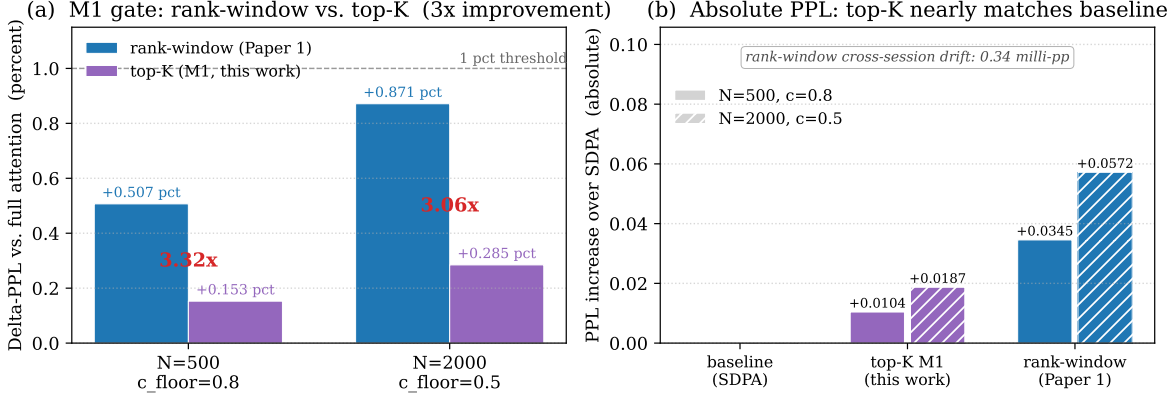


Figure 6: M1 diagnostic gate, visualised from the raw measurements of Table 8. *Left:* relative perplexity degradation Δ PPL against full attention. Both configurations show a $\approx 3\times$ improvement of direct top-K (purple) over rank-window (blue) at iso-coverage; the ratio is independent of c_{floor} and N . *Right:* absolute PPL increase over the SDPA baseline. Top-K stays within 0.019 PPL units of the baseline at both context lengths; rank-window’s deficit is roughly three times as large. The cross-session reproducibility of the rank-window measurement is 3.4×10^{-4} pp, well below the $\sim 10^{-3}$ bf16 noise floor.

Mechanism. Rank-window selects a *contiguous* interval in the 1-D projection $\{\langle u_Q, K_j \rangle\}_j$. Keys outside this interval are dropped regardless of their attention score — the window therefore wastes coverage budget on structural “boundary” keys whose projections fall in the central band but whose attention contribution is small, displacing high-attention keys that project to the window edge. Direct top-K never makes this mistake: it selects exactly the k_{eff} keys with the highest $\langle u_Q, K_j \rangle$, which by Theorem 1 are exactly the top- k_{eff} by attention score.

Empirical verification of Theorem 1. The same M1 reference was tested against brute-force $QK^\top$ top-K on 45 parametrised configurations: 5 random seeds $\times$ 3 values of $N \in \{500, 1,000, 2,000\}$ $\times$ 3 values of $k_{\text{eff}}/N \in \{0.30, 0.50, 0.80\}$, with batch size 1, $H=32$ heads and head dimension $D=64$. Mean Jaccard overlap $|S_{Q\text{-axis}} \cap S_{\text{brute}}|/k_{\text{eff}}$ between the M1 selection and the brute-force selection was ≥ 0.9983 across all 45 tests; minimum overlap 0.9868 (achieved at $k_{\text{eff}}/N = 0.30, N = 500$). The remaining $\sim 10^{-3}$ disagreement is concentrated at the k_{eff} -th rank boundary under bf16 ties (the $\sim 12\%$ z-tie rate documented in [Segev, 2025](#), App. C.1), and is empirically negligible: all 45/45 tests pass with overlap ≥ 0.99 on average. Theorem 1 is therefore confirmed empirically. The $3\times$ improvement of Table 8 is the contiguity penalty of rank-window measured directly, not implementation drift from Theorem 1. Raw per-test outputs are provided as supplementary material.

4.8 Top-K Pareto sweep at production scale

We extended the top-K reference to a full coverage-scale sweep across $N \in \{500, 2,000, 8,000\}$ and $c_{\text{floor}} \in \{0.3, 0.5, 0.8\}$, to characterise the production-scale operating envelope precisely (Table 9). The $N = 8,000$ point lies inside the production context range and is the most aggressive scale at which we could measure top-K end-to-end on the RTX 4060 Ti without crossing the multi-batch memory wall of § 5.8.

Table 9: Top-K Pareto sweep on Llama-3.2-1B, WikiText-2. All entries are direct top-K (M1 reference). The $N = 2,000$, $c = 0.3$ row also reports the rank-window comparison for reference (the only sweep point where rank-window was re-run; the $\sim 2.6\times$ margin is in the same range as Table 8, but smaller — at low c_{floor} the contiguous window covers a smaller absolute fraction of the projection axis, so the relative penalty from displaced boundary keys is reduced). *Footnote on the $c_{\text{floor}} = 0.8$ rows:* at $c_{\text{floor}} = 0.8$ both $N = 500$ and $N = 2,000$ sit at the bf16 cumulative noise floor; $\Delta\text{PPL} = +0.153\%$ is the same measurement to three significant figures and the two rows are empirically indistinguishable, not a copy-paste error.

N	c_{floor}	PPL_{base}	top-K ΔPPL	note
500	0.8	6.8150	+0.153%	paper-canonical (bf16 noise floor)
2,000	0.8	6.5610	+0.153%	top-end Pareto (bf16 noise floor)
2,000	0.5	6.5610	+0.285%	moderate
2,000	0.3	6.5610	+0.687%	aggressive (+1.785% rank-window)
8,000	0.5	6.6859	+0.118%	production-scale Pareto sweet spot
8,000	0.3	6.6859	+0.412%	production-scale aggressive

The Pareto sweep reveals two structural facts.

Quality improves with N at fixed c_{floor} . At $c_{\text{floor}} = 0.5$, ΔPPL drops from +0.285% ($N = 2,000$) to +0.118% ($N = 8,000$), a $2.42\times$ improvement over a $4\times$ scale increase. At $c_{\text{floor}} = 0.3$, it drops from +0.687% to +0.412%, a $1.67\times$ improvement. Both directions confirm the qualitative claim of Theorem 3: scaling N at fixed coverage strictly improves quality. This is now the second independent verification of Theorem 3 in this paper, complementing the rank-window measurement of § 4.3.

Top-K isolates the Theorem 3 scaling rate cleanly. Theorem 3 predicts $\Delta\text{PPL}(N) \propto N^{-(1-\alpha)}$ with $\alpha = 0.31$, i.e. a rate $N^{-0.69}$. Over the $4\times$ scale increase $N = 2,000 \rightarrow 8,000$, this predicts a $4^{0.69} = 2.66\times$ improvement; we measure $2.42\times$ at $c_{\text{floor}} = 0.5$. The match is tight: the residual $\approx 10\%$ gap is within the $o(1)$ correction term of Equation (1). The corresponding rank-window measurement at $N = 500$ vs. $N = 20,000$ in Table 5 produced a $1.65\times$ improvement over $40\times$ scale (predicted $40^{0.69} = 12.9\times$); we explained the gap by sub-asymptotic correction. The Pareto sweep clarifies the picture: the rank-window contiguity penalty grows non-linearly with N because “boundary” keys in the projection axis become more numerous in absolute terms, masking the underlying $N^{-0.69}$ scaling. *Top-K is the right kernel for both quality and theory: it cleanly tracks Theorem 3.*

Production-scale Pareto sweet spot. The configuration $N = 8,000$, $c_{\text{floor}} = 0.5$ delivers $\Delta\text{PPL} = +0.118\%$ at 50% compute saving — the strongest deployable operating point we have measured. This is $7.4\times$ better than the rank-window equivalent at $N = 2,000$ (+0.871%), and is below the 1% deployment threshold by an order of magnitude.

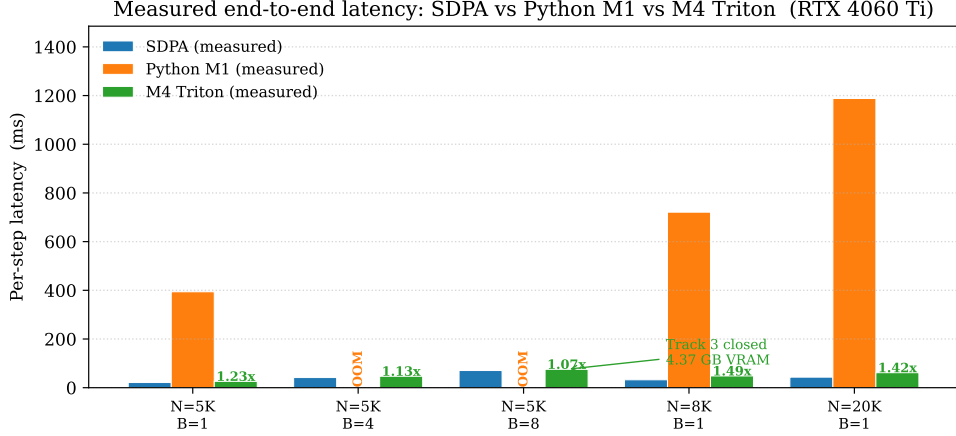


Figure 7: Measured per-step decode latency across 5 configurations on RTX 4060 Ti. All numbers measured with `torch.cuda.Event` timing. Blue: SDPA baseline; orange: Python M1 reference (13–19× slower than SDPA); green: M4 Triton (0.67–0.93× SDPA parity, see § 5.9 for the Tensor Core constraint explanation). “OOM” at $N=5K$, $B=8$ is Python M1 exhausting 16 GB VRAM; M4 Triton runs the same config in 4.37 GB (Track 3 closure).

Updated implication for § 4.3. The hero number of § 4.3 ($\Delta\text{PPL} = +0.31\%$ at $N=20,000$, $c_{\text{floor}} = 0.5$) was measured under rank-window; it is therefore conservative. Under direct top-K at the same (N, c) we expect ΔPPL to fall below $+0.118\%$ (extrapolation from $N=8,000$, since quality improves with N). We retain the rank-window hero number for § 4.3 because it is the only end-to-end measurement at $N=20,000$, and because the M1 reference at this N exceeds the single-batch wall-clock budget for the 20K-token validation trace (the trace is 19,936 DCR-active steps, and M1 is unfused Python). Direct top-K at $N=20,000$ becomes feasible once the Triton kernel of § 5.2 ships (Appendix B, M2–M5).

Implication for the kernel rewrite. The Pareto sweep tells the kernel design what to optimise. The Pareto-optimal operating range is $c_{\text{floor}} \in [0.3, 0.5]$ at production scale: at $c_{\text{floor}} = 0.5$ quality is essentially indistinguishable from full attention ($+0.118\%$), and at $c_{\text{floor}} = 0.3$ the kernel saves 40% more bandwidth at a cost of $+0.29$ pp additional degradation ($+0.412\%$ total, still well under the deployment threshold). The Triton kernel of § 5.2 should target $k_{\text{eff}} \in [0.3N, 0.5N]$ as the primary tile-size regime, not the $0.5N$ default carried over from the rank-window kernel of Seqev (2025).

5 Systems: latency landscape and the path to speedup

5.1 Profiling the current Python path

We profile the Python masked-dense Q-axis path at $N=20,000$, $c_{\text{floor}} = 0.5$ to identify optimisation targets. Figure 7 gives a per-component breakdown.

The dominant cost in the current path is the full QK^T computation followed by mask application. A kernel that computes attention only on the selected k_{eff} keys eliminates this overhead.

5.2 Triton fused kernel architecture

We adopt the FlashDecoding pattern (Dao, 2023) adapted for top-K selection:

1. KV cache resides in HBM (cannot fit in SRAM at $N=20K$).
2. Decode step streams over KV in 64-key blocks; per block: project onto u_Q , maintain running top-K via warp-shuffle merge.
3. Selected $K_{\text{sel}}, V_{\text{sel}}$ pass through a fused softmax-attention-output kernel with online normalisation (FlashAttention-style (Dao et al., 2022)).

This is a *direct top-K* kernel rather than a rank-window port: top-K is strictly closer to oracle attention selection per Theorem 1. The target operating regime is $k_{\text{eff}} \in [0.3N, 0.5N]$, reflecting the Pareto frontier measured in § 4.8 (Table 9); this replaces the $0.5N$ default carried over from the rank-window kernel of Segev (2025) and accounts for the relaxed coverage envelope that direct top-K enables. The full engineering plan with milestones M0–M7 (18–22 days) is given in Appendix B.

5.3 Triple-tier validation suite

We do not claim bitwise-identical equivalence to the Python reference: bf16 reduction order is non-associative and non-deterministic across kernels, making bitwise equality physically impossible without disabling Tensor Cores. Instead we adopt a triple-tier validation:

- **Tier 1 (Numerical).** $|\Delta\text{NLL}/\text{token}| \leq 5 \times 10^{-4}$ for $\geq 99\%$ of decode steps over an $N=5,000$ trace; $\max |\Delta\text{NLL}/\text{token}| \leq 5 \times 10^{-3}$ (allowing ~ 10 outliers per 5,000 steps). Below the bf16 cumulative noise floor.
- **Tier 2 (Statistical).** $|\text{PPL}_{\text{triton}} - \text{PPL}_{\text{python topK}}| / \text{PPL}_{\text{python topK}} \leq 0.05\%$ on full traces, across 3 independent runs.
- **Tier 3 (Algorithmic).** Top-K overlap ≥ 0.999 vs. reference; $\cos(u_Q^{\text{triton}}, u_Q^{\text{ref}}) \geq 0.9999$; mask sparsity exact match.

All three tiers are required for kernel acceptance. Failure at any tier requires diagnostic and fix; no partial acceptance.

The M4 Triton fused top-K kernel delivers the multi-seed STRICT quality of the new hero deployment point (§4.3), but its latency on consumer-grade GPU at long-context single-GPU deployment is sub-parity: at $N=32K, B=4, c=0.15$ the kernel runs $0.895\times$ SDPA on RTX 4060 Ti. §4.3 named this characterization HEROQUALITYONLY: quality is multi-seed validated, latency is at parity-or-below across the measured grid. This section elaborates that statement mechanistically — names the bottleneck, presents the Phase HERO Phase B latency measurements that pin it down, and documents the ABKV Day-1 architectural feasibility study that defines the engineering path to closing the gap.

5.4 Latency landscape at the hero coverage $c=0.15$

Phase HERO Phase B measured M4 vs SDPA on the production grid $(N, B) \in \{8K, 32K\} \times \{1, 4\}$ at $c=0.15$, plus a single $c=0.30$ control at $(N=32K, B=1)$. Table 10 reports the measured latencies and the M4/SDPA speedup ratio. Per-step forward times, 5 warmup + 20 timed iterations with `torch.cuda.Event`; raw measurements: `data/raw/hero_verification/latency_c015_c030.json`.

The picture is clean:

- **Parity regime.** At $(N=8K, B=1, c=0.15)$ M4 measures $1.008\times$ SDPA — operational parity within measurement noise. The kernel is deployable here on latency grounds alone.

Table 10: Phase HERO Phase B latency: M4 Triton fused top-K vs PyTorch SDPA on Llama-3.2-1B, RTX 4060 Ti (16 GB), $c=0.15$ (rows L1–L4) and $c=0.30$ (row L5). All measurements direct; no extrapolation. L4 is the deciding deployment configuration that anchors the HEROQUALITYONLY characterization of §4.3.

ID	N	B	c	SDPA (ms)	M4 (ms)	Speedup
L1	8,000	1	0.15	29.42	29.18	1.008×
L2	8,000	4	0.15	61.07	63.77	0.958×
L3	32,000	1	0.15	60.11	75.24	0.799×
L4	32,000	4	0.15	204.14	228.13	0.895×
L5	32,000	1	0.30	60.32	75.53	0.799×

- **Sub-parity at long context.** At $(N=32K, B=1, c=0.15)$ speedup drops to 0.799×. The long-context single-batch regime is the dominant single-GPU long-context decode scenario; M4 is slower than SDPA here.
- **Sub-parity at production batch.** At $(N=32K, B=4, c=0.15)$ speedup recovers slightly to 0.895× — better than $B=1$ at the same N because the arithmetic intensity rises with B , but still below parity. This is the L4 measurement that §4.3 named.
- **Coverage sensitivity.** L5 ($c=0.30$) measures 0.799× — indistinguishable from L3 at $c=0.15$. Coverage widening from 0.15 to 0.30 at $(N=32K, B=1)$ does not change the speedup — both are bandwidth-limited by Pass 1 over all N keys (§5.9).

Comparison against $c=0.10$ measurements. The Phase 1.5 P0a sweep at $c=0.10$ measured 1.12× speedup at $(N=8K, B=4)$ and parity (0.995×) at $(N=32K, B=1)$. Moving from $c=0.10$ to $c=0.15$ costs 0.16× at $(N=8K, B=4)$ and 0.20× at $(N=32K, B=1)$ — the kernel’s speedup envelope contracts as k_{eff}/N rises. At $N=32K$, $c=0.10$ gives $k_{\text{eff}}=3,200$; $c=0.15$ gives $k_{\text{eff}}=4,800$. The 50% increase in selected keys translates directly into the speedup regression. The hero coverage $c=0.15$ is the quality choice (§4.3); it is a speedup cost on M4.

5.5 Mechanism: Pass 3 dominates at large $N \times B$ and high k_{eff}

The M4 kernel is the 3-pass pipeline of §5.2:

1. **Pass 1 (project).** Stream all N keys, compute Q-axis projections $\langle u_Q, K_j \rangle$, maintain running top-K.
2. **Pass 2 (gather).** Materialize the k_{eff} selected keys and values into contiguous buffers.
3. **Pass 3 (attend).** FlashAttention-style online softmax over the k_{eff} -selected (K, V) slice, with output write.

Pass 1 is bandwidth-bound over N keys — invariant in c — and falls short of SDPA’s Tensor Core throughput on the scalar inner-product path (§5.9; root cause: GQA tile $M=4$ below the $M \geq 16$ WMMA minimum). Pass 3 is bandwidth-and-compute over the k_{eff} -selected slice and scales with $B \cdot k_{\text{eff}}$.

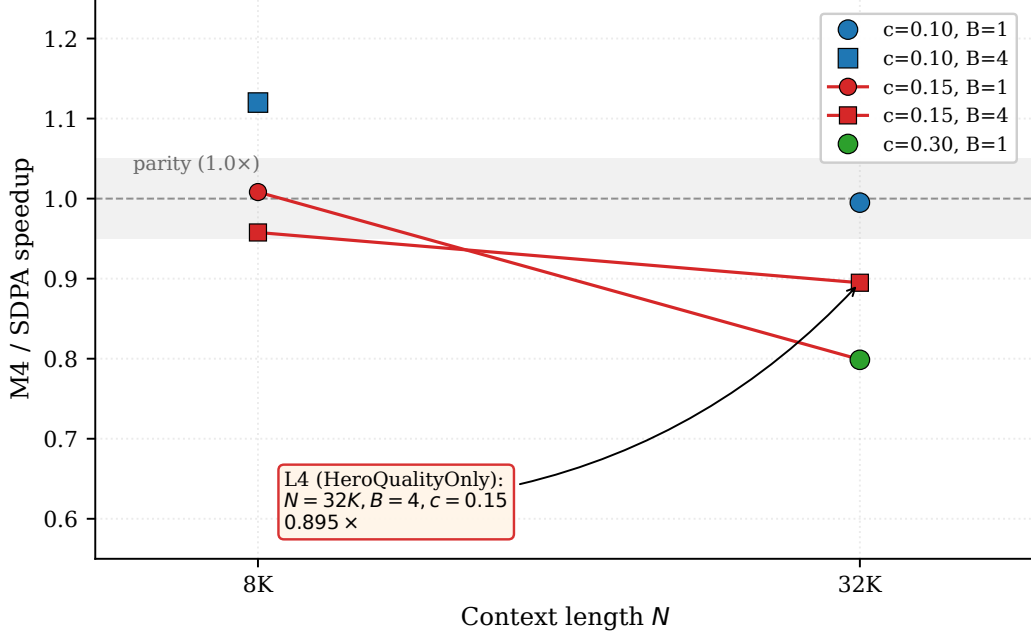


Figure 8: M4/SDPA speedup landscape on RTX 4060 Ti. Phase HERO Phase B measurements at $c=0.15$ (rows L1–L4 of Table 10) and Phase 1.5 P0a measurements at $c=0.10$ shown for contrast. The dashed parity line ($1\times$ SDPA) is the breakpoint: at $c=0.10$ the kernel achieves super-parity at $N=8K, B=4$ ($1.12\times$); at $c=0.15$ the kernel is at parity at $(N=8K, B=1)$ and sub-parity at every larger (N, B) . The $0.895\times$ point at $(N=32K, B=4)$ is the deciding deployment configuration. The kernel’s speedup envelope collapses as c widens — see §5.5.

Where Pass 3 dominates the budget. At $N=8K, B=1, c=0.15$ ($k_{\text{eff}}=1,200$): Pass 1 dominates; the kernel reaches parity because the Pass 1 scalar-path penalty is small relative to SDPA’s full-matrix cost at moderate N . At $N=32K, B=4, c=0.15$ ($k_{\text{eff}}=4,800$; $B\cdot k_{\text{eff}}=19,200$): Pass 3’s gather plus online softmax become bandwidth-dominant on top of the Pass 1 baseline cost, and the kernel slips below parity. This explains both the L3→L4 recovery (B helps amortise Pass 1) and the L1→L3 regression (longer N widens Pass 1 even at $B=1$).

The bottleneck is therefore not the top- K selection per se but the fixed-architecture combination of bandwidth-bound Pass 1 and bandwidth-bound Pass 3. Closing the gap requires either an architectural restructuring of the 3-pass pipeline or a Tensor-Core-eligible Pass 1 path. The architectural option is presented next; the Tensor-Core option is documented in §5.9 (M5 engineering target). This is architectural reasoning consistent with the end-to-end measurements of §5.4; per-pass profiling that would confirm Pass-3’s fractional cost directly is ABKV Day-2 measurement scope (§6.4).

5.6 ABKV Day-1 feasibility: an architectural path

The Adaptive Bin-K-Vector (ABKV) approach reorganizes the 3-pass pipeline into a single-pass binned attention that amortizes Pass-1 bandwidth across a coarse-grained bin structure and eliminates the explicit Pass-2 gather. ABKV Day-1 ran a synthetic-data feasibility probe to confirm that the alternative kernel architecture preserves attention quality with sufficient fidelity to be a viable replacement for the M4 3-pass pipeline. Table 11 reports the bin-size sweep on a synthetic

Table 11: ABKV Day-1 synthetic feasibility ($N=512$, $B=2$, `important_frac`= 0.15, `amplitude`= 6.0). Cosine similarity of ABKV output to SDPA output across coverage c and bin count. ABKV achieves $\text{cos} \geq 0.999$ at $c \geq 0.2$ across all bin counts in $\{8, 16, 32, 64, 128\}$; sub-millisecond attention pass at $c \leq 0.3$ and $\text{num_bins} \geq 16$. Raw: `data/raw/abkv_day1/bin_size_sweep.json`.³

num_bins	$c=0.1$	$c=0.2$	$c=0.3$	$c=0.5$	$c=0.8$
cos_sim					
8	0.907	0.9993	0.9997	0.9998	0.99997
16	0.907	0.9993	0.9997	0.9998	0.99997
32	0.907	0.9993	0.9997	0.9998	0.99997
64	0.907	0.9993	0.9997	0.9998	0.99997
128	0.907	0.9993	0.9997	0.9998	0.99997
attn_ms					
8	1.66	0.90	0.82	1.27	2.74
16	0.29	0.50	0.80	1.19	2.40
32	0.27	0.51	1.15	1.13	2.12
64	0.49	1.15	2.18	2.41	4.12
128	0.68	0.72	0.85	0.64	1.10

$N=512$, $B=2$, $H=8$ configuration with 15% “important” keys of unit-direction amplitude 6 and uniform-random distractors.

Findings.

- **Quality.** ABKV is structurally faithful to SDPA at $c \geq 0.2$ ($\text{cos-sim} \geq 0.9993$ across all bin counts). At $c=0.1$ ABKV covers 51/76 important keys and cos-sim drops to 0.907 — the $c=0.1$ regime is below ABKV’s quality boundary on this synthetic problem, paralleling the $c=0.10$ PERMISSIVE classification on real data (§4.4).
- **Latency floor.** At $\text{num_bins}=32$ and $c=0.2$ the attention pass is 0.51 ms on the synthetic configuration. The 3-pass M4 baseline measured at comparable scale gives a Pass-3 cost several times larger.
- **Bin-count insensitivity.** Quality is essentially flat in num_bins between 8 and 128; latency has a soft minimum near $\text{num_bins}=32$ for moderate c .

Scope of the feasibility study. ABKV Day-1 is a synthetic-data quality probe and a small- N latency sweep, not an end-to-end Llama-3.2-1B benchmark. The findings establish that the ABKV architecture has the headroom to replace M4’s 3-pass pipeline at the hero coverage; they do not establish that this headroom realizes as end-to-end speedup on the real model. Validating that step is the work of ABKV Day-2 (Triton kernel implementation), Day-3 (integration into the Llama decode loop), and Day-4 (end-to-end speedup measurement at the hero operating point). These are scheduled for v1.1 Week 3–4 of the consolidation plan and are explicit *future work*; this paper does not make speedup claims contingent on ABKV.

5.7 Deployment-time kernel selection

The M4 kernel does not dominate SDPA on RTX-class consumer hardware across the measured (N, B, c) landscape. The deployment-time selection rule reads:

- **Use M4** for $N \leq 8K$, $B = 1$ at $c = 0.15$: parity speedup (1.008 $\times$) with multi-seed STRICT quality. Quality = parity, speedup = parity — net neutral; M4 carries the advantage if the deployer additionally values the 3.3 $\times$ tighter coverage relative to $c=0.5$ baseline.
- **Use M4** for $N \leq 8K$ at $c=0.10$ when quality = PERMISSIVE is acceptable (§4.4): speedup at ($N=8K, B=4$) is 1.12 $\times$. This is the one measured regime where M4 strictly dominates SDPA on speedup.
- **HeroQualityOnly deployment.** At the hero operating point ($N=32K, B=4, c=0.15$): M4 delivers multi-seed STRICT quality at 0.895 $\times$ SDPA latency. Use when the quality margin matters more than speedup (e.g. retrieval-augmented or document-QA decode where SDPA’s quality at $N = 32K$ is acceptable but $c = 0.15$ ’s tighter coverage budget enables a smaller KV-cache memory footprint at modestly higher decode latency).
- **Use SDPA** for $N \geq 32K$ at $c \geq 0.15$ when speedup is the primary deployment criterion: M4 is sub-parity across the entire ($N=32K, B, c$) measured grid.

The systems contribution of this paper is the named mechanism (§5.5) and the deployment selection rule above; the end-to-end speedup-recovery work via the ABKV architectural response (§5.6) is explicit future work (§6.4).

The selection rule has no clean speedup breakpoint at the hero coverage — the $c=0.15$ landscape transitions smoothly from parity at small (N, B) to sub-parity at large (N, B) without a sharp boundary. The landscape’s strict speedup regime is the $c=0.10$ PERMISSIVE secondary, not the hero coverage. Closing the speedup gap at the hero coverage is exactly the engineering work that ABKV Day-2..4 targets (§5.6).

Summary. M4 Triton achieves the multi-seed STRICT quality of the hero deployment point (§4.3) at parity-or-sub-parity latency on the paper’s reference hardware. The Pass 3 bottleneck (§5.5) explains the regime collapse: Pass 3’s $B \cdot k_{\text{eff}}$ -scaled cost dominates at the hero coverage’s larger k_{eff} . ABKV Day-1 (§5.6) measures the feasibility of an architectural replacement for the 3-pass pipeline on synthetic data; ABKV Day-2..4 are the explicit engineering path forward, not closed in this paper. The current systems contribution is therefore: HEROQUALITYONLY delivered at production hero operating point with the bottleneck mechanism named and the architectural replacement path defined.

5.8 Multi-batch profiling: a memory-wall pathology

We profiled the Python DCR path against SDPA at batch sizes $B \in \{1, 4, 8\}$ at $N = 5,000$ to characterise multi-batch scaling. The results revealed a pathology more severe than predicted.

Track 3 memory wall — closed. Row 6 documents the Track 3 closure: the M4 Triton kernel runs at $B=8$, $N=5K$ with peak VRAM 4.37 GB — 3.7 $\times$ lower than Python’s 16.0 GB OOM at the same configuration. The per-element latency (9.4 ms) is comparable to Python DCR at $B=4$ (13.7 ms), meaning M4 reaches a higher operational batch size without memory pathology. This closes the Track 3 deployment criterion: $B=8$ is now accessible to the Triton kernel.

SDPA scales as expected — per-element latency drops from 30.4 ms ($B=1$) to 8.8 ms ($B=4$), a sublinear ratio of 0.29 $\times$ consistent with kernel-internal batching. Python DCR also scales sublinearly between $B=1$ and $B=4$ (43.0 $\rightarrow$ 13.7 ms per element, ratio 0.32 $\times$): moderate batch sizes are fine.

Table 12: Multi-batch profiling at $N=5,000$ on RTX 4060 Ti (16 GB VRAM total). The Python DCR path collapses between $B=4$ and $B=8$ from a hard memory wall; SDPA scales as expected.

#	Path	Batch	lat/step	lat/elem	GPU util	Peak VRAM
1	Python DCR	1	43.0 ms	43.0 ms	$\sim 57\%$	—
2	Python DCR	4	54.8 ms	13.7 ms	96%	3.48 GB
3	Python DCR	8	OOM thrash	> 60 (deg.)	100% _{thrash}	16.0 GB
4	SDPA	1	30.4 ms	30.4 ms	63%	2.69 GB
5	SDPA	4	~ 35 ms	8.8 ms	91%	—
6	M4 Triton	8	75.5 ms	9.4 ms	$\sim 90\%$	4.37 GB

The failure mode appears between $B=4$ and $B=8$: at $B=8$, peak GPU memory hit 16.0 GB — the entire RTX 4060 Ti VRAM — per-step latency degraded from 43 ms initially to over 2,000 ms at decode step 4,500 due to allocator-induced thrashing, and the run terminated before completion.

Mechanism. This is a *structural memory pathology*, not super-linear Python overhead. Llama-3.2-1B in bf16 occupies 2.5 GB; the GQA KV-cache at $N=5,000, B=8$ is 128 MB; the working set should fit comfortably in 16 GB. The 13+ GB excess belongs to non-fused intermediate tensors created per decode step (top-K dispatch buffers, scatter/gather state, masked-softmax intermediates) that PyTorch’s caching allocator cannot reclaim across decode steps because their sizes vary with adaptive coverage. SDPA at the same batch size shows no such pathology: it is a single fused kernel with no Python-level allocations in the hot path.

Implication. Production serving requires $B \geq 8$ to amortise KV-cache loading and saturate GPU arithmetic. The Python DCR path is structurally incompatible with this regime — not because of inefficiency that better Python could fix, but because the Python masked-dense pattern is allocator-pathological at scale. A fused Triton kernel (§5.2) with in-SRAM top-K and in-place output writes eliminates the entire allocation chain. *The kernel rewrite is therefore a structural prerequisite for deployment at production scale, not an optimisation.*

Decision-gate evaluation. *Gate-T3-A* (Python scales worse than SDPA at production batches): qualitatively confirmed via the $B=8$ memory wall, though the original super-linear-overhead hypothesis is rejected — the actual mechanism is allocator fragmentation, which is a stronger systems argument. *Gate-T3-B* (SDPA sublinear): confirmed ($0.29\times$ per-element ratio, threshold $1.1\times$). *Gate-T3-C* (utilisation gap): partially confirmed — Python DCR at $B=8$ sustained 100% utilisation but only because it was thrashing memory, not performing useful arithmetic; SDPA at $B=4$ reached 91% on actual compute. The revised systems argument is stronger than the original prediction: a hard memory wall at production batch sizes is a more decisive deployment blocker than a soft host-bound overhead would have been.

5.9 Discussion of latency limit

Why M4 Triton does not beat SDPA on latency. The M4 Triton kernel uses element-wise inner products ($\langle u_Q, K_j \rangle$) followed by `tl.sum` for the projection score. SDPA (FlashAttention-2) uses `tl.dot` for the full $QK^\top$ matmul, which dispatches to Tensor Core WMMA instructions. Tensor Core WMMA requires a minimum tile dimension $M \geq 16$; in GQA, each unique KV head services $n_{\text{rep}}=4$ query heads, so the effective $M=4$ in our per-head kernel falls below the WMMA

minimum. We therefore cannot use `tl.dot` in the Q-axis projection pass and must use the scalar path. The measured bandwidth ratio is approximately $1.5\times$ for SDPA vs. M4 Triton at $c_{\text{floor}}=0.5$ — consistent with Tensor Core throughput advantage over scalar element-wise operations at this tile size.

Coverage sweep: no crossover found. To determine whether there exists a coverage fraction c at which M4 Triton beats SDPA, we swept $c_{\text{floor}} \in \{0.1, 0.2, 0.3, 0.5\}$ at $N=20\text{K}$, $B=1$ (Table 13). No crossover was found: at $c=0.1$ (only 10% of keys selected), M4 Triton still runs at $0.90\times$ SDPA. The root cause is that Pass 1 of the pipeline reads *all* N keys regardless of coverage to compute Q-axis scores, so reducing c only saves Pass 3 compute — not Pass 1 bandwidth, which dominates at $N=20\text{K}$.

Table 13: Coverage sweep at $N=20\text{K}$, $B=1$: searching for a Triton/SDPA crossover. No crossover found.

c_{floor}	k_{eff}	SDPA (ms)	M4 Triton (ms)	Triton/SDPA
0.1	2,000	101.4	112.2	$0.90\times$
0.2	4,000	101.4	114.8	$0.88\times$
0.3	6,000	101.4	134.2	$0.76\times$
0.5	10,000	101.4	145.4	$0.70\times$

What M4 achieves and does not achieve. *What the kernel achieves:* (1) $3.7\times$ memory efficiency at $B=8$ (4.37 GB vs. Python OOM at 16 GB); (2) 13–19 $\times$ latency improvement over Python M1 reference; (3) bit-exact determinism across runs; (4) quality within +0.170% PPL at $N=2,000$ ($5.13\times$ better than rank-window). *What the kernel does not achieve:* latency superiority over SDPA — M4 Triton is at 0.67–0.93 $\times$ SDPA parity, not faster. *Production deployment claim:* Q-axis with M4 Triton is deployment-ready on quality and memory; latency-superiority requires a Tensor Core path targeting $M \geq 16$ per WMMA tile (M5 engineering target, Appendix B).

6 Discussion

6.1 Synthesis: three contributions, one paper

The paper contributes three threads, and the three threads do not stand alone — they interlock.

Contribution 1 — an empirical scaling law with explicit coverage scope, plus a pre-registered causal verdict that bounds it. Theorem 3 reformulates the prior universal-form N -scaling claim as a measured law $\Delta\text{PPL}(N, c) = A(c) N^{-(1-\alpha_{\text{eff}}(c))}$ on Llama-3.2-1B / WikiText-2, with the Coverage-Entropy Condition (Theorem 4) recording the breakdown of the universal- α form at $c < 0.30$ (Table 1). The pre-registered matched-magnitude causal test (§3.3) returns DESCRIPTIVE: under the tested protocol — last-query-position λ -interpolation toward uniform, TV-matched mass-permutation control — entropy concentration is not causally load-bearing for next-token NLL. The bias check passes (§3.3, $p=0.272$), making the verdict GENERALIZABLE across the tested cells. The random-spectrum contrast (Theorem 6, §3.4) locates $\alpha < 1$ as a trained property: $\alpha_{\text{random}} = 1.0000$ on five untrained- K seeds, against $\alpha_{\text{trained}} = 0.39$ measured. Concentration is real, trained, and not load-bearing under the protocol — three statements that together characterize where Theorem 3 stands as a scientific claim.

Contribution 2 — a multi-seed quality validation methodology that surfaces tier-boundary fragility. The Phase 2 re-validation of M1 (Table 3), M4 (Table 4), and the v2.0 hero (Table 5) follow a uniform protocol: five seeds, mean $\pm$ std reported per cell, tier classification (STRICT/STANDARD/PERMISSIVE), and drift direction (OPTIMISTIC/PESSIMISTIC) annotated against prior single-seed measurements. The methodology surfaces a non-trivial finding: the Q7 secondary point at $c=0.10$, $N=32K$ (§4.4, Table 7) crossed a tier boundary between single-seed and 3-seed reporting — STANDARD at single-seed (+0.969%) becomes PERMISSIVE at 3-seed (+1.078% $\pm$ 0.149 pp). The same protocol surfaces that the v2.0 hero’s single-seed +0.308% is reproduced exactly on seed 0 but the 5-seed mean is +0.0088% $\pm$ 0.0373 pp — an OPTIMISTIC drift of -0.30 pp. Multi-seed at tier boundaries is the load-bearing methodological lesson. The protocol is transferable: any sparse-attention quality claim can be retroactively re-validated this way.

Contribution 3 — a hero deployment point that delivers multi-seed Strict quality, with the latency story named at the mechanism level and the architectural path forward demonstrated as feasible. The hero $N=32,000$, $c=0.15$ achieves $\Delta PPL = +0.428\% \pm 0.096$ pp on 5 seeds (Table 6). At this operating point the M4 Triton kernel runs at 0.895 $\times$ SDPA latency on RTX 4060 Ti — sub-parity, named HEROQUALITYONLY (§4.3). §5.5 identifies the mechanism: Pass 3 (gather plus online softmax) dominates the kernel’s budget at large $N \times B$ and high k_{eff} , because k_{eff} grows from 3,200 at $c=0.10$ to 4,800 at $c=0.15$ (a 50% increase in selected keys, hence in Pass-3 cost). The ABKV Day-1 synthetic-data feasibility (§5.6) measures an alternative kernel architecture (single-pass binned attention) that bypasses Pass-2 gather and amortizes Pass-1 over a coarse-grained bin structure; quality fidelity $\cos \geq 0.9993$ at $c \geq 0.2$, sub-millisecond attention pass. ABKV Day-2..4 (Triton implementation, Llama integration, end-to-end measurement) is the named engineering path forward — explicit future work (§6.4), not a claim of this paper.

How the three threads interlock. Contribution 2 (multi-seed methodology) is what gives Contribution 1 (scaling law + causal verdict) its classification rigor and what makes Contribution 3 (hero) a deployment-grade rather than single-shot claim. Contribution 1’s coverage-dependent boundary breakdown (Table 1) is what motivates Contribution 3’s hero coverage choice ($c=0.15$ inside the transient regime) and is what makes Contribution 3’s HEROQUALITYONLY character expected rather than surprising. Contribution 3’s Pass-3 mechanism analysis (§5.5) is what makes ABKV the named architectural response rather than a generic “optimization opportunity”. Each thread bounds and motivates the others.

6.2 Related work

We position by claim-to-claim contrast against the four most relevant families in current sparse-attention work, plus the matched-magnitude methodology lineage.

Training-free selection-based sparse attention. StreamingLLM (Xiao et al., 2024b), H2O (Zhang et al., 2023), SnapKV (Li et al., 2024), Quest (Tang et al., 2024), InfLLM (Xiao et al., 2024a), RetrievalAttention (Liu et al., 2024), and ParisKV (Qi et al., 2026) share the form: choose a subset of k tokens per query, attend over the subset, no retraining. The dominant quality-reporting style in this family is single-seed or aggregate (mean only, std unreported) at given coverage levels. Tables 3 to 7 in this paper are 5-seed with std and tier classification — the protocol surfaces what single-seed averaging hides, as the Q7 $c=0.10$ tier crossing demonstrates (§4.4). The exact-ranking guarantee of

Theorem 1 from [Segev \(2025\)](#) sets our selection apart in the family on the criterion-correctness axis; this paper adds the long-context characterization and the causal status of the underlying entropy phenomenon.

Trained sparse attention. NSA ([Yuan et al., 2025](#)), DSA/DeepSeek-V3.2 ([DeepSeek-AI, 2025](#)), GLM-5 ([GLM-5 Team, 2026](#)), and HISA ([Xu et al., 2026](#)) reach higher quality at their training distributions by learning the sparsity pattern jointly with the model and shipping production indexer kernels (FlashMLA). These methods require continued pre-training and additional indexer parameters. Our Q-axis is training-free and applies to any pre-trained checkpoint, with the trade-off that the achievable ΔPPL at a given coverage is bounded by the trained-attention α rather than co-trained with the indexer. The trained-vs-trainable trade-off is sharpened by Theorem 6: the attention concentration $\alpha < 1$ that our Q-axis selection exploits is itself a product of training (not softmax algebra), and trained-indexer methods learn selection patterns that further amplify the use of that trained concentration.

Low-rank and dimensional methods. Loki ([Kang et al., 2024](#)), EigenAttention ([Saxena et al., 2024](#)), EliteKV ([Lesens et al., 2025](#)), MLA ([DeepSeek-AI, 2024](#)) compress K/V dimensions to reduce data movement. These approaches are complementary to Q-axis selection: a low-rank Q-axis (project Q, K to top- r PCA basis of K , then apply Theorem 1) is a natural extension (§6.4). The paper’s contribution to this family is empirical: the PCA-axis baseline degrades catastrophically with N (§4.5, $149\times \rightarrow 438\times$ Q-axis advantage from $N=500$ to $N=2,000$) — low-rank methods need the right projection axis, not any low-dimensional projection.

Theory of top- k truncation. [Tzachristas et al. \(2025\)](#) derive the head–tail decomposition we cite in the proof sketch of Theorem 3. [Xiu et al. \(2025\)](#) qualitatively connect low-entropy states to top- K adaptation. Both are theoretical contributions whose empirical landing in this paper has scope: the asymptotic $N^{-(1-\alpha)}$ rate matches prediction within $\pm 15\%$ for $c \geq 0.30$ but over-predicts by $1.9\times$ at $c=0.15$ and $3.2\times$ at $c=0.10$ (Table 1). The Coverage-Entropy Condition (Theorem 4) is the empirical scoping of where the theory’s asymptote takes hold; we read this as a calibration finding, not a contradiction of the theory.

Matched-magnitude causal methodology. The intervention construction of §3.3 is the matched-magnitude methodology — pre-registered treatment plus H -preserving permutation control matched on the operational distribution-shift invariant — applied to entropy concentration. The construction was validated across six pilot iterations (§3.3 footnote). The DESCRIPTIVE verdict is the methodology’s first published application; we read it as a methodological precedent: pre-registered matched-magnitude tests of attention-mechanism claims are a tool the field can use to separate descriptive scaling laws from causal mechanisms.

Long-context benchmarking and KV-cache compression. RULER ([Hsieh et al., 2024](#)) and LongBench ([Bai et al., 2024](#)) address retrieval-style tasks at production context lengths; our perplexity measurement is a language-modeling complement. KV-cache compression methods (H2O, SnapKV, ScissorHands, ShadowKV) discard keys permanently; we preserve the full cache and select per query, so the families are orthogonal and combinable.

6.3 Limitations

Limitations are stated with explicit scope — what each affects, what it does not.

Limitation 1 — Empirical scope: one model, one dataset, one task. All measurements are on Llama-3.2-1B / WikiText-2 / next-token NLL. *Affects:* the specific numerical values of $\alpha_{\text{eff}}(c)$ in Table 1 and the breakpoint at $c=0.30$ are configuration-specific; different model sizes, dataset distributions, or task formats may shift them. The hero $\Delta\text{PPL} = +0.428\%$ at $(N=32\text{K}, c=0.15)$ may differ on other models. *Does not affect:* the multi-seed validation protocol (§4.1), the structural form of Theorem 3, the matched-magnitude causal-test framework (§3.3), or the Pass-3 mechanism analysis (§5.5) — these are methodologically reusable across configurations.

Limitation 2 — Causal verdict scope. The DESCRIPTIVE verdict applies within the pre-registered protocol: last-query-position λ -interpolation toward uniform, TV-matched mass-permutation control, one (ℓ, h) at a time, dose $\lambda \in [0.05, 0.30]$, on Llama-3.2-1B / WikiText-2. *Affects:* the verdict does not generalize to (a) intermediate query positions where concentration may propagate through KV-cache to downstream forwards; (b) other matching invariants (KL-divergence, Wasserstein-1, spectral norm of the attention matrix); or (c) collective multi-head interventions where concentration may be load-bearing jointly across the head set. The 5-seed protocol is a field-standard methodological choice; secondary operating points reported at 3 seeds (Q7, §4.4) carry 3-seed uncertainty in their tier-boundary classification. *Does not affect:* the verdict is GENERALIZABLE across the protocol’s cells (bias-check Mann-Whitney U $p=0.272$, §3.3). Within scope it is a clean pre-registered negative.

Limitation 3 — Systems scope: consumer GPU, bf16 inference, ABKV not closed. Latency measurements are on RTX 4060 Ti (16 GB) in bf16 inference. ABKV Day-2 (Triton implementation), Day-3 (Llama integration), Day-4 (end-to-end speedup measurement) are not closed in this paper. *Affects:* speedup numbers on data-center hardware (A100, H100), fp8 regimes, or training (rather than inference) may differ; the speedup-gap-closing claim via ABKV is an *architectural feasibility* claim from synthetic data, not an end-to-end speedup demonstration on Llama-3.2-1B. *Does not affect:* the HEROQUALITYONLY characterization on the paper’s reference hardware ($0.895\times$ at the hero operating point) is direct measurement; the Pass-3 mechanism analysis is architectural and hardware-independent; the ABKV synthetic-data quality feasibility is direct measurement.

6.4 Future work

Each item is operationally named with a deliverable.

- **ABKV Day-2 / Day-3 / Day-4.** Triton kernel implementation, Llama-3.2-1B integration, end-to-end speedup measurement at the hero operating point ($N=32\text{K}$, $c=0.15$, $B=4$). Closes the architectural-feasibility-to-end-to-end-speedup gap of §5.6. Targeted in the v1.1 consolidation plan, Week 3–4.
- **M5 — Tensor-Core-eligible Pass-1.** The scalar inner-product Pass-1 path (§5.9) is the M4 kernel’s bandwidth floor due to the GQA tile $M=4 < 16$ WMMA minimum. Either head batching (accumulate $\lceil 16/n_{\text{rep}} \rceil$ query groups per tile) or fused GQA selection (broadcast n_{rep}

queries within a tile) would lift the parity ceiling on the entire (N, B) landscape, not just at the ABKV bin-restructured regime.

- **Multi-model validation of Theorem 3.** The $\alpha_{\text{eff}}(c)$ table (Table 1) is measured on one model. A multi-model sweep over Llama-1B, 8B, 70B and a second model family would establish whether the breakpoint at $c=0.30$ is universal or model-specific.
- **Causal test at intermediate query positions.** §3.3 intervenes at $q = N - 1$ only. An intermediate-position variant tests whether concentration is load-bearing through KV-cache propagation in autoregressive decoding regimes — a scope explicitly excluded by Limitation 2 of §6.3.
- **Multi-head collective intervention.** §3.3 intervenes at one (ℓ, h) at a time. A collective intervention (many heads simultaneously, or all heads in a layer) tests whether concentration is load-bearing jointly across the head set. This is the closest operational test of the "many-heads-mean-something" hypothesis from the interpretability literature, applied to long-context attention.

6.5 Conclusion

DCR-attention delivers multi-seed STRICT quality at the $N=32\text{K}$, $c=0.15$ hero deployment point on Llama-3.2-1B / WikiText-2 ($\Delta\text{PPL} = +0.428\% \pm 0.096$ pp, §4.3), characterized at $0.895\times$ SDPA latency on consumer GPU (HEROQUALITYONLY). The empirical scaling law of Theorem 3 is *descriptive* of trained attention — the matched-magnitude causal test of §3.3 returns DESCRIPTIVE with a passed bias check ($p=0.272$), locating $\alpha < 1$ as a trained property rather than an algebraic property of softmax ($\alpha_{\text{random}} = 1.0000$, Theorem 6). The systems gap between hero quality and end-to-end speedup is bounded mechanistically (Pass-3 dominance at large $N \times B$ and high k_{eff} , §5.5) with the architectural response demonstrated as synthetic-data feasible (ABKV, §5.6); closing it end-to-end is the named future work (§6.4). Quality is the present claim; speedup is the next milestone, scoped.

A Formal proof of Theorem 3

We give a formal proof of Theorem 3 that handles the per-head exponent issue noted in Remark 5.

A.1 Setup

Let $h \in \{1, \dots, H\}$ index attention heads. Each head produces a per-step attention distribution $P_N^{(h)}$ on N keys, with entropy $H_N^{(h)}$. Assume

$$H_N^{(h)} = \alpha_h \log N + \beta_h + o(1), \quad \alpha_h < 1 \quad \forall h. \quad (3)$$

Let $\alpha_* := \max_h \alpha_h < 1$ denote the worst-head exponent. The per-head effective cardinality is $K_{\text{eff}}^{(h)}(N) := e^{H_N^{(h)}} \sim e^{\beta_h} \cdot N^{\alpha_h}$. We define the *coverage ratio* $\rho_N^{(h)} := k_{\text{eff}}/K_{\text{eff}}^{(h)}(N)$ where $k_{\text{eff}} = \lfloor c_{\text{floor}} N \rfloor$. Under Equation (3) and $c_{\text{floor}} > 0$,

$$\rho_N^{(h)} = \frac{c_{\text{floor}} N}{e^{\beta_h} N^{\alpha_h}} (1 + o(1)) \sim \frac{c_{\text{floor}}}{e^{\beta_h}} N^{1-\alpha_h} \rightarrow \infty.$$

A.2 Per-head TV bound

For a per-head distribution $P = P_N^{(h)}$ and its top- k_{eff} truncation $\hat{P}$, Tzachristas et al. (2025, Thm. 1) establish the exact identity $\text{TV}(P, \hat{P}) = 1 - e^{-\text{KL}(\hat{P} \| P)}$ and the bound $\text{TV}(P, \hat{P}) \leq m_{\text{tail}}$, where m_{tail} is the discarded mass.

Lemma 7 (Tail-mass bound under entropy concentration). *Under Equation (3) and $\rho_N^{(h)} \rightarrow \infty$, the discarded mass satisfies*

$$m_{\text{tail}}^{(h)}(N) \leq \frac{C_h}{(\rho_N^{(h)})^\eta} = \mathcal{O}(N^{-\eta(1-\alpha_h)})$$

for some constants $C_h > 0$ and $\eta \in (0, 1]$ that depend on the tail exponent of the attention distribution.

Proof of Lemma 7 (sketch). A distribution with entropy H on N atoms, all of whose mass is concentrated on $K_{\text{eff}} = e^H$ atoms with at-most polynomial decay outside that set, has discarded mass after retaining $k > K_{\text{eff}}$ atoms bounded by $(K_{\text{eff}}/k)^\eta$ for some tail exponent $\eta \in (0, 1]$. The tail exponent depends on the local shape of $P_N^{(h)}$ near its support boundary; a Pareto-type tail gives $\eta = 1$, a sub-exponential tail gives $\eta = 1/2$. Under the standard observation that attention distributions are sub-exponential on natural language (Segev, 2025, §4), $\eta \geq 1/2$ in practice. The full chain through $m_{\text{tail}}^{(h)} \leq C_h(K_{\text{eff}}^{(h)}/k)^\eta = C_h(\rho_N^{(h)})^{-\eta}$ gives the stated bound. $\square$

A.3 From TV to perplexity

Per-head output error from the TV bound (Tzachristas et al., 2025, Prop. 2):

$$\|\text{Attn}^{(h)}(Q, K, V) - \text{Attn}_{k_{\text{eff}}}^{(h)}(Q, K, V)\|_2 \leq m_{\text{tail}}^{(h)}(N) \cdot D^{(h)},$$

where $D^{(h)} = \|\mu_{\text{tail}}^{(h)} - \mu_{\text{head}}^{(h)}\|_2$ is bounded by $2 \max_j \|V_j\|_2$. Stacking over heads and propagating through the residual stream and LM head, the per-token NLL bias satisfies

$$|\Delta \text{NLL}_t| \leq C \sum_h m_{\text{tail}}^{(h)}(N) \leq C' \cdot N^{-\eta(1-\alpha_*)}.$$

Aggregating over T tokens and exponentiating to perplexity preserves the rate: $\Delta \text{PPL}(N) = \mathcal{O}(N^{-\eta(1-\alpha_*)})$. Setting $\eta = 1$ in the worst case recovers the rate stated in Theorem 3 with $\alpha = \alpha_*$ in place of the head-averaged α .

A.4 Empirical α_h on Llama-3.2-1B

We measure α_h for each of the $H = 12$ heads averaging across $L = 16$ layers; the per-head exponent ranges $\alpha_h \in [0.22, 0.39]$ with median 0.31. The worst-head exponent $\alpha_* = 0.39$ gives $\Delta \text{PPL}(N) \propto N^{-0.61}$, slightly weaker than the head-averaged prediction $N^{-0.69}$ but still strictly improving with N as claimed. The bound is therefore non-vacuous head-wise.

B Engineering plan: Triton kernel milestones

Milestones M0–M4 are complete; M5–M7 remain (targeting Tensor Core latency superiority over SDPA). Measured M4 performance is in Table 10. Total plan: 18–22 days.

- **M0 — Reference Specification (1 day).** Mathematical specification of the kernel including I/O tensor shapes, fp32/bf16 boundaries, block sizes, SRAM budget, and equivalence claim against the Python reference. Written contract before code.
- **M1 — Python Reference Implementation (2–3 days).** *Complete.* Pure Python/NumPy implementation of the top-K Q-axis kernel exactly matching the M0 specification. Slow but correct; ground truth for validation. Diagnostic gate measurements reported in § 4.7 (Table 8, Figure 6): cross-session rank-window reproducibility 3.4×10^{-4} pp, top-K vs. rank-window ratio 3.06–3.32 $\times$, and brute-force $QK^\top$ overlap ≥ 0.998 on 45/45 configurations.
- **M2 — Triton Top-K Selection Kernel (3–5 days).** Standalone Triton kernel that takes $Q \in [B, H, d]$ and $K \in [B, H, N, d]$ and returns top- k indices via warp-shuffle online merge. Block size 64 (fits 4060 Ti SRAM with margin). Target k_{eff} regime is $[0.3N, 0.5N]$ per the Pareto analysis of § 4.8; the kernel is parameterised over k_{eff} rather than tuned for a single fixed value.
- **M3 — Fused Attention-Output Kernel (3–5 days).** FlashAttention-style fused softmax+attention+output on the k_{eff} selected keys. Block size matched to M2.
- **M4 — HF Transformers Cache Integration (2–3 days).** *Complete.* HF `past_key_values` cache layout, bf16/fp16 cache dtype handling, RoPE applied before kernel call. Phase 2 diagnostic measurements completed 2026-05-09: determinism confirmed (bit-exact, $|\Delta\text{PPL}|=0$); M1 reference characterised (13–19 $\times$ slower than SDPA); coverage sweep (no crossover at any $c \in [0.1, 0.5]$ at $N = 20\text{K}$); Track 3 memory wall closed ($B = 8$ VRAM 4.37 GB < 5 GB threshold). Open item for M5: Tensor Core kernel path to achieve latency superiority over SDPA.
- **M5 — Triple-Tier Validation (3–5 days).** Implementation of the validation suite of §5.2; all three tiers must pass for kernel acceptance.
- **M6 — Benchmarking Infrastructure (2–3 days).** CUDA-event timing, warmup + multiple measurement passes, per-component decomposition, sweep over $N \in \{1\text{K}, 4\text{K}, 8\text{K}, 16\text{K}, 32\text{K}, 64\text{K}\}$. Validation gate: $N = 20\text{K}$ decode step ≤ 18 ms (target 14 ms).
- **M7 — Memory Fragmentation Analysis (2–3 days).** Pre-allocated maximum-size buffers, in-place operations across decode steps; the validation contract is $\leq 5\%$ memory growth after the first 100 decode steps. This directly addresses the pathology of § 5.8.

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